Human-Centered Artificial Intelligence (HCAI)

Heart Disease Prediction System

Abstract

This report presents a Human-Centered AI (HCAI) based Heart Disease Prediction System implemented as a Streamlit web application. The project combines machine learning algorithms with an interactive user interface to assist users in identifying possible heart disease risks. The system emphasizes human oversight, transparency, ethical AI principles, and healthcare accessibility. Multiple machine learning algorithms were evaluated and compared, and the Random Forest Classifier was selected as the final deployed model due to its superior performance. The project aligns with Sustainable Development Goals related to health, education, and innovation.

Introduction

Heart disease is one of the leading causes of death worldwide. Early risk identification can help individuals seek preventive healthcare and improve medical outcomes. This project demonstrates how machine learning and Human-Centered AI principles can be applied to healthcare support systems.

The application is designed around the principles of Human-Centered AI:

- Transparency
- User control
- Ethical AI usage
- Human oversight
- Accessibility

The system collects patient health metrics, preprocesses the input data, and uses a trained machine learning model to predict the probability of heart disease risk. The prediction is advisory in nature and is intended to support awareness rather than replace professional medical diagnosis.

Project Overview

Project Name: Heart Disease Prediction System
Implementation: Python Streamlit application with machine learning based heart disease prediction.

Primary Functions

- Collect patient health indicators through a web interface.
- Preprocess user inputs for machine learning inference.
- Predict heart disease risk using a trained classifier.
- Display risk probability and risk category using visual indicators.
- Support healthcare awareness through understandable outputs.

Target Users

- Students learning Machine Learning and HCAI concepts.
- Healthcare support practitioners.
- Users interested in basic heart disease risk awareness.

Technologies Used

Technology	**Purpose**
Python	Core programming language
Pandas	Data preprocessing and handling
Scikit-learn	Machine learning algorithms
Streamlit	Frontend web application
Matplotlib	Data visualization
Seaborn	Correlation heatmaps and plots
LabelEncoder	Categorical data encoding

Machine Learning Algorithms Used

The following machine learning algorithms were implemented and compared:

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbour (KNN)

4. Random Forest Classifier

The algorithms were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

The Random Forest Classifier achieved the highest accuracy and was selected as the final model for deployment.

Human and Computer Automation Levels

This system follows a collaborative automation model with strong human oversight.

Human Role

- Enters patient health information.
- Initiates prediction manually.
- Reviews risk probability and prediction results.
- Makes the final healthcare decisions.

Computer Role

- Processes user inputs.
- Executes machine learning inference.
- Predicts heart disease risk.
- Displays probability and risk level.

Automation Level (HCAI Perspective)

- Human-led interaction and decision-making.
- Computer-assisted prediction execution.
- Advisory output instead of autonomous decision-making.

The project therefore operates in a human-in-the-loop framework where AI supports analysis while humans retain control.

SDG Mapping

SDG 3: Good Health and Well-being

- Supports early detection of heart disease risks.
- Encourages preventive healthcare awareness.
- Promotes healthcare accessibility through AI systems.

SDG 4: Quality Education

- Demonstrates practical applications of machine learning.
- Supports educational learning in AI and HCAI.
- Provides an interactive healthcare AI learning platform.

SDG 9: Industry, Innovation and Infrastructure

- Showcases digital healthcare innovation using accessible web-based AI.
- Encourages reproducible deployment of machine learning models.
- Demonstrates lightweight healthcare infrastructure using AI.

SDG 10: Reduced Inequalities

- Provides accessible healthcare support tools.
- Uses a simple and understandable user interface.
- Promotes healthcare technology accessibility for non-technical users.

Implementation Artifacts

Frontend Application

The system frontend was developed using Streamlit and includes:

- Interactive patient detail forms
- Healthcare-focused dashboard interface
- Prediction result cards
- Risk probability metrics
- Demo patient profiles
- Color-coded risk indicators

Implementation Artifacts

Frontend Application Interface

The frontend application was developed using Streamlit with a healthcare-oriented dashboard design. The interface allows users to enter patient health information such as age, cholesterol level, blood pressure, heart rate, and exercise angina status. The system also includes demo patient profiles for testing different healthcare scenarios.

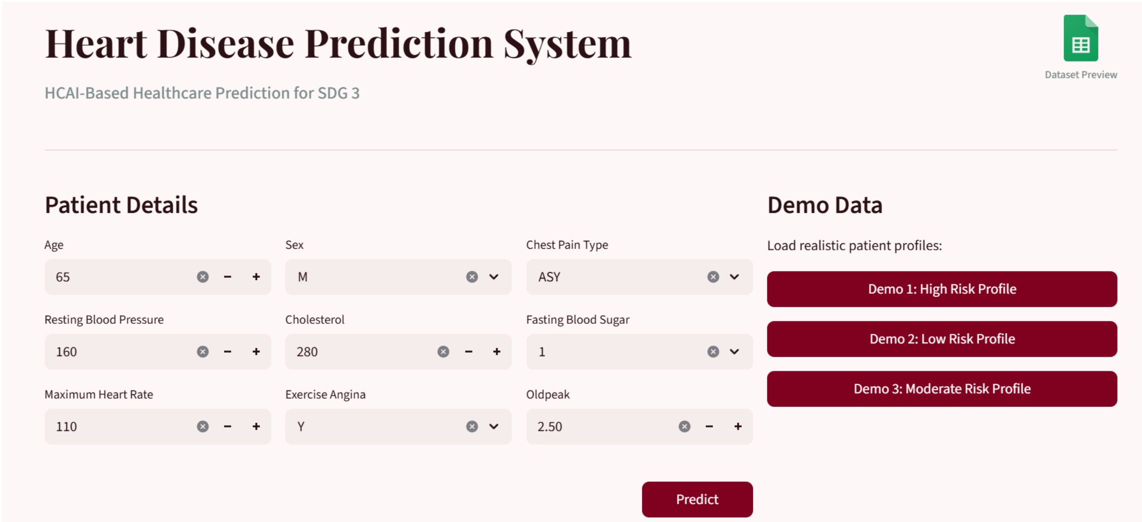

Figure 1: Frontend Interface of the Heart Disease Prediction System

High Risk Prediction Output

When the entered patient metrics indicate a high probability of heart disease, the system displays a high-risk warning message along with the calculated probability score and healthcare recommendation. The interface uses color-coded visual indicators to improve readability and risk awareness.

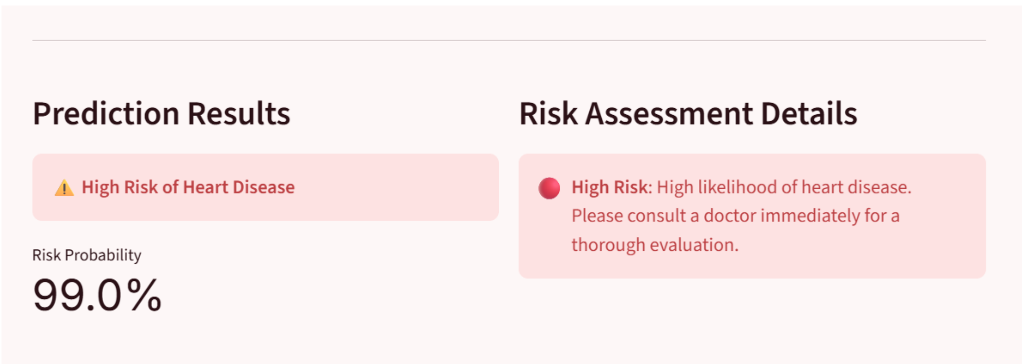

Figure 2: High Risk Heart Disease Prediction Output

Low Risk Prediction Output

When the patient metrics indicate a low probability of heart disease, the system displays a low-risk prediction with positive healthcare guidance and risk probability information.

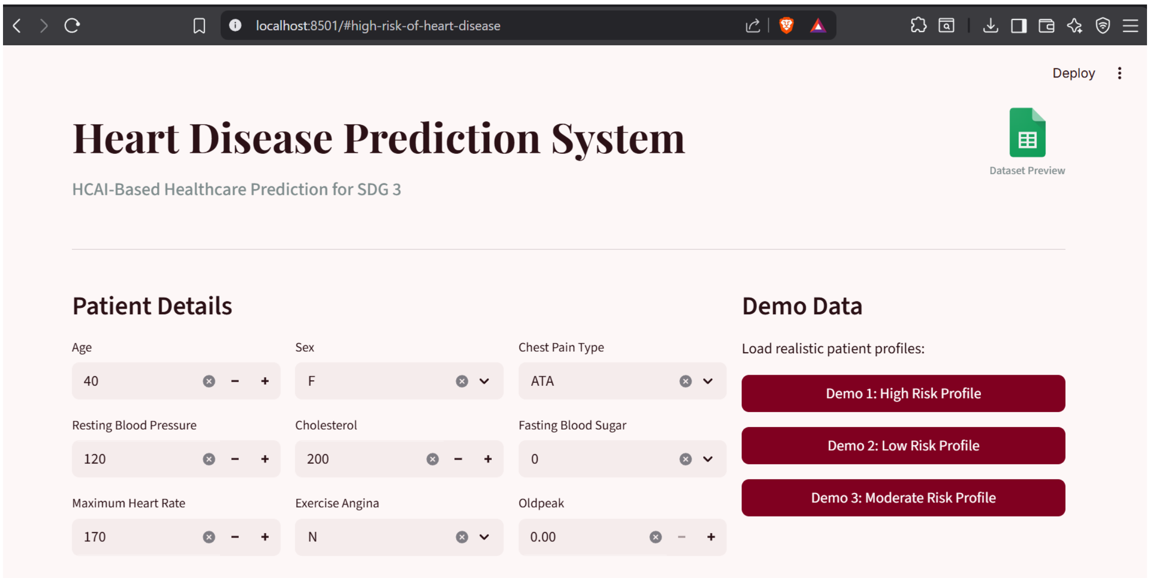

Figure 3: Low Risk Patient Input Example

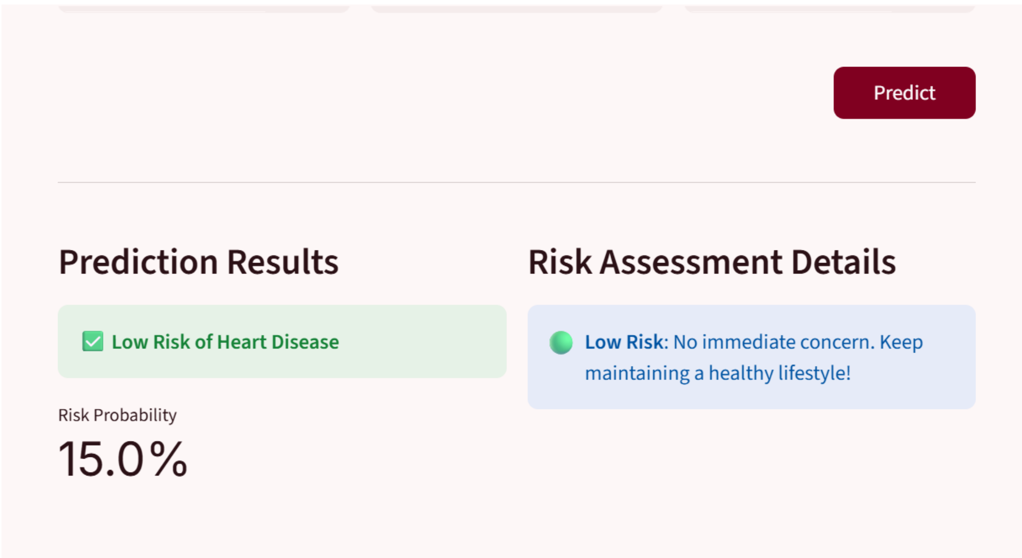

Figure 4: Low Risk Heart Disease Prediction Output

The frontend emphasizes Human-Centered AI principles by providing:

- User-friendly interaction
- Clear healthcare guidance
- Visual risk indicators
- Human-controlled prediction execution

- Explainable prediction outputs

Key Features

- Responsive web interface
- Real-time prediction
- User-friendly healthcare dashboard
- Risk probability visualization
- Demo profiles for testing

Detailed System Workflow

1. Dataset Preparation

- Heart Disease dataset was collected and preprocessed.
- Categorical data was encoded using LabelEncoder.
- Numerical features were normalized where required.

2. Data Visualization

The dataset was analyzed using:

- Histograms
- Boxplots
- Correlation heatmaps
- Pair plots

3. Model Training

- The dataset was divided into training and testing sets.
- Multiple algorithms were trained and evaluated.
- Performance metrics were compared.

4. Frontend Prediction

- Users enter patient details.
- Input data is processed by the Random Forest model.
- Prediction and probability score are displayed.

Design Metaphors

Clinical Assistant

The application behaves like a lightweight healthcare assistant that helps users understand possible heart disease risks.

Risk Radar

The prediction interface acts like a risk radar that visually highlights possible health risks.

Healthcare Dashboard

The system interface resembles a healthcare dashboard with organized patient metrics and risk indicators.

Decision Support System

The application supports human healthcare decisions without replacing medical professionals.

Ethics and Responsible AI

Transparency

- The system clearly states that predictions are advisory.
- Risk probability is displayed openly.

Human Oversight

- Humans remain responsible for healthcare decisions.
- The AI system does not replace doctors.

Privacy

- No persistent patient data is stored.
- The application avoids collecting personally identifiable information.

Fairness and Bias Awareness

- Predictions depend on the quality of training data.
- Model bias and dataset limitations are acknowledged.

Accessibility Features

- Simple user interface for non-technical users.
- Clearly labeled input fields.
- Color-coded prediction indicators.
- Interactive and responsive design.

HCAI Best Practices Embedded

- Human control over prediction execution.
- Explainable risk outputs.
- Advisory rather than autonomous AI behavior.
- Ethical AI boundaries and disclaimers.
- Interactive user feedback.

Recommendations for Future Improvements

- Add explainable AI visualizations.
- Improve model interpretability.
- Add healthcare chatbot support.
- Expand dataset diversity.
- Include multilingual accessibility features.

Conclusion

This Heart Disease Prediction System demonstrates how Human-Centered AI can support healthcare awareness while maintaining human oversight and ethical AI principles. The project combines machine learning, healthcare analytics, and accessible frontend design to create an educational and socially beneficial healthcare support system aligned with Sustainable Development Goals.